

incompressibleDriftFlux — Exact Solution Algorithm

Traced directly from OpenFOAM-12 source (applications/modules/{incompressibleDriftFlux, twoPhaseSolver, VoFSolver}), as configured for the IWA mini-hydrocyclone sludge/water case (Journal_Run/final_coarse).

1. Physical model

Two incompressible phases — sludge (dispersed) and water (continuous) — are modelled as a single mixture with one shared velocity field `U`. The relative (slip) motion between the phases is not solved as a separate momentum equation; instead it is closed algebraically via a drift velocity model. This is the Euler-Euler mixture / algebraic-slip (drift-flux) approach, as opposed to a full two-fluid Euler-Euler formulation.

2. Fields and auxiliary quantities

Symbol	Field	Role
$\alpha_1 = \text{alphad}$	sludge volume fraction	solved (transport equation)
$\alpha_2 = \text{alphac} = 1 - \alpha_1$	water volume fraction	algebraic
<code>U</code>	mixture velocity	solved (momentum equation)
<code>p_rgh</code> , <code>p</code>	pressure minus hydrostatic; true pressure	solved / reconstructed
$\rho = \alpha_1 \rho_d + \alpha_2 \rho_c$	mixture density	algebraic, updated every step
$\nu = \mu(\rho_c \nu_c, U) / \rho$	mixture kinematic viscosity	<code>slurry::mu()</code> correlation
<code>U_dm</code>	drift (slip) velocity of sludge relative to mixture	<code>relativeVelocityModels::general</code>
<code>k</code> , <code>ε</code> (or <code>ω</code>)	turbulence closure quantities	RAS transport equations

3. Per-time-step algorithm (PIMPLE outer-corrector loop)

Step 1 — prePredictor()

incompressibleDriftFlux.C:179

(a) Advect the phase fraction

Combined mean + drift flux, bounded MULES solve:

$$\partial\alpha_1/\partial t + \nabla \cdot [(\varphi + \nabla \cdot \varphi(U_{dm})) \cdot \alpha_1] = 0$$

(MULES-bounded, $0 \leq \alpha_1 \leq 1$)

$\varphi(U_{dm})$ is the extra flux contributed by the drift velocity — this term is the entire mechanism by which sludge and water physically separate in the hydrocyclone.

(b) Extra explicit turbulent diffusion correction

Solved separately (this is the "alpha.sludge Diffusion" entry in fvSolution):

$$\partial\alpha_1/\partial t - \nabla \cdot (v_t \nabla \alpha_1) = 0$$

(c) Update mixture properties

incompressibleDriftFluxMixture.C:87–91

$$\rho = \alpha_1 \rho_d + \alpha_2 \rho_c$$

$$v = \mu_{\text{mix}}(\rho_c \cdot v_c, U) / \rho$$

$$\mu_{\text{mix}} = \rho_c \cdot v_c \cdot (1 + 2.5\alpha_1 + 10.05\alpha_1^2 + 0.00273 \cdot e^{16.6\alpha_1}) \quad [\text{slurry model, Thomas correlation}]$$

(d) Mass flux

$$\rho\varphi = \varphi_{\alpha 1} \cdot \rho_d + \varphi_{\alpha 2} \cdot \rho_c$$

(e) Update the drift velocity

general.C:66–76 — algebraic-slip / equilibrium-Eulerian closure:

$$\mathbf{a} = \mathbf{g} - (\mathbf{U} \cdot \nabla) \mathbf{U} \quad (\text{mixture acceleration: gravity + convective})$$

$$U_{dm} = (\rho_c / \rho) \cdot V_c \cdot \mathbf{a} \cdot [e^{-a \cdot \max(\alpha_1 - \alpha_{res}, 0)} - e^{-a1 \cdot \max(\alpha_1 - \alpha_{res}, 0)}]$$

V_c carries dimensions of **time** — it is a particle response/relaxation time scale, not a velocity. This is the classic algebraic-slip closure: slip velocity $\approx$ (particle relaxation time) $\times$ (local acceleration) $\times$ (concentration-dependent hindrance function).

(f) Turbulence predictor

`momentumTransport->predict()` — explicit part of the k/ϵ (or k/ω) transport equations.

Step 2 — `momentumPredictor()`

`VoFSolver/momentumPredictor.C:33–69`

$$\partial(\rho U)/\partial t + \nabla \cdot (\rho \phi U) + \text{MRF terms} + \nabla \cdot \tau_{\text{dev}} = f_\sigma + \nabla \cdot \tau_{\text{Dm}}$$

solved (implicitly, via the assembled matrix) for U , with:

$$\nabla \cdot \tau_{\text{dev}} = \text{momentumTransport->divDevTau}(U) \quad (\text{viscous} + \text{Reynolds stress, uses } v_{\text{Eff}} = v + v_t)$$

$$\begin{aligned} \nabla \cdot \tau_{\text{Dm}} &= \text{relativeVelocity->divDevTau}() \quad (\text{extra "turbulent dispersion" stress from phase slip}) \\ &= \nabla \cdot (\alpha_d \rho_d \cdot U_{\text{dm}} \otimes U_{\text{dm}} + \alpha_c \rho_c \cdot U_{\text{cm}} \otimes U_{\text{cm}}) \end{aligned}$$

Step 3 — `pressureCorrector()` : PISO loop

`twoPhaseSolver/pressureCorrector.C:41–147`

$$H_{\text{byA}} = r_{\text{AU}} \cdot H(U) \quad (r_{\text{AU}} = 1/A(U), \text{ inverse of the momentum matrix diagonal})$$

$$\phi_{\text{HbyA}} = \text{flux}(H_{\text{byA}}) + \text{ddtCorr}(U, \phi)$$

$$\phi_g = (\text{surface tension} - g \cdot h \cdot \nabla \rho) \cdot r_{\text{AU}}$$

$$\nabla \cdot (\phi_{\text{HbyA}} + \phi_g) - \nabla \cdot (r_{\text{AU}} \nabla p_{\text{rgh}}) = 0 \quad (\text{pressure equation, solved for } p_{\text{rgh}})$$

$$\phi = \phi_{\text{HbyA}} + \phi_g + (\text{flux correction from } p_{\text{rgh}} \text{ solve})$$

$$U = H_{\text{byA}} + r_{\text{AU}} \cdot \text{reconstruct}(\phi_g + p_{\text{rgh}} \text{ flux correction})$$

$$p = p_{\text{rgh}} + \rho \cdot g \cdot h \quad (\text{recover true pressure})$$

Repeated `nCorrectors` times (PISO inner loop) per outer PIMPLE iteration.

Step 4 — postCorrector()

`momentumTransport->correct()` — implicit corrector solve of the RAS closure equations (for `kEpsilon`: the k and ϵ transport equations), using the just-updated $\mathbf{U}$, $\mathbf{v}$, and production $\mathbf{G}$ computed from $\nabla \mathbf{U}$.

Outer loop: Steps 1–4 repeat for `nOuterCorrectors` PIMPLE iterations per timestep (transient run, `ddtSchemes: Euler`, `adjustTimeStep on` with Courant-number control via `maxCo`), then time is advanced and the whole sequence repeats.

4. What this reduces to for your current setup

With `alpha.sludge  $\equiv 0$` everywhere (no sludge fed at either inlet):

- Step (a)'s drift flux is zero — nothing to advect differently from a passive scalar at 0.
- Step (e): `$\max(\alpha_1 - \alpha_{\text{res}}, 0)$` = 0 $\Rightarrow$ both exponentials evaluate to 1 $\Rightarrow$ the bracket is exactly zero $\Rightarrow \mathbf{U}_{\text{dm}} = \mathbf{0}$ everywhere, for all time.
- Consequently $\nabla \cdot \tau_{\text{Dm}} = 0$ in the momentum equation (Step 2).
- Step (c)'s viscosity does *not* reduce exactly to ν_{water} : at $\alpha_1=0$ the slurry correlation gives $\mu_{\text{mix}} = \mu_c \times 1.00273$ — a small, constant, domain-wide +0.273% viscosity offset baked into the formula's constant term.

Net effect: the case runs as an essentially single-phase incompressible RAS momentum/pressure solve, with molecular viscosity ~0.27% higher than pure water — negligible next to the differences between turbulence models being compared, and identical across all 16 cases (so it does not bias the comparison).